TAREK SHEREEN DARWISH

Backend Software Engineer | APIs, Cloud & Data

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Open to remote roles, or relocation with visa sponsorship

SUMMARY

Backend-focused engineer finishing a B.Sc. in Intelligent Systems Engineering (2027). Built a FastAPI fraud-detection service (test ROC AUC 0.97), a Node.js and TypeScript API with JWT auth on MySQL and Redis, and the Supabase row-level-security model for a health platform that won the YLF Hackathon 2026. Merged a librespot (Rust) fix. Seeking a junior backend role.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL, Rust, C

APIs: FastAPI, Node.js/Express, REST API design, JWT auth, Postman, pytest

Data: PostgreSQL (Supabase) with row-level security, MySQL, Redis, Firebase

Cloud & DevOps: Docker, Docker Compose, GitHub Actions, Azure, Cloudflare Pages, Hugging Face Spaces

EXPERIENCE

Generative AI Engineer (DEPI Generative AI Track), Digital Egypt Pioneers Initiative — *Cairo, Egypt (Hybrid) · Nov 2025 – Jul 2026*

- Built an explainable fraud-detection service: XGBoost on 310 features (test ROC AUC 0.97, precision 0.89, recall 0.74), served by FastAPI with pytest tests in GitHub Actions.
- Used Gemini to turn top SHAP factors into plain-English explanations (template fallback); Dockerized on Hugging Face Spaces.

Backend & AI Engineer Intern, Aispeak — *Madrid, Spain (Remote) · Jul 2025 – Jan 2026*

- Built the backend pipeline that routes user messages to Azure and a LLaMA-70B model.
- Set up real-time messaging on Supabase and owned the database schema and API design.

Backend Engineer Intern, Jiffy — *Remote · Aug 2025 – Dec 2025*

- Built the Node.js and TypeScript backend (Express, MySQL, Redis, Docker Compose) with JWT auth and admin and user endpoints.
- Wrote Postman collections for dev, staging and production, plus setup scripts and docs.

Software Engineer Intern, PETROBEL — *Nasr City, Egypt · Sep 2024 – Oct 2024*

- Supported a legacy .NET to modern .NET (MVC) migration that cut system load time by 20%.

PROJECTS

Healthica: Smart Band Health Platform · *Winner, YLF Hackathon 2026 (team of 5) · 2nd place, Banha · Supabase, React, FastAPI, Flutter*

- Designed the Supabase schema and row-level security for four roles: doctors read only their assigned patients' data, using security-definer helpers to avoid policy recursion.
- Built the React and TypeScript doctor and admin dashboard on Cloudflare Pages, with Supabase Realtime for live alerts.
- Served the TFLite glucose models through a FastAPI gateway (REST, WebSocket, MQTT); also built the Flutter app.

safi: Local Agentic LLM Backend · *Python, llama.cpp · in progress*

- Local backend built around a planner and an executor model on llama.cpp. Done so far: a model probe that measures VRAM, speed and thermals on a 16 GB laptop GPU, a tool registry and a CPU embedder, with pytest tests.

Risk AI: Industrial IoT Safety Monitor · *Docker, SQLite, ESP32, React*

- Dockerized backend storing ESP32 readings every 2 s and predicting gas-leak risk from trends; React and Recharts dashboard.

OPEN SOURCE

- Merged: librespot (Rust), #1786, fixes Liked Songs updates sent as plain JSON being dropped as invalid base64.
- Open: Sonora (Rust), Spotify Connect, Jams, right-to-left text; MeasureIt_ARCH (Python), #309.

EDUCATION

B.Sc., Intelligent Systems Engineering, Helwan National University — *Helwan, Egypt · 2022 – 2027 (expected)*

CERTIFICATIONS

AWS Cloud Practitioner · Huawei HCIP-AI · Huawei ICT Associate · ML Specialization (Stanford)